42.7: Adaptations for gas exchange include pigments that bind and transport gases

The high metabolic demands of many animals necessitate the exchange of large quantities of O_2 and CO_2. Here we'll examine how blood molecules called respiratory pigments facilitate this exchange through their interaction with O_2 and CO_2. We'll also investigate physiological adaptations that enable animals to be active under conditions of high metabolic load or very limiting P_{O_2}. As a basis for exploring these topics, let's summarize the basic gas exchange circuit in humans.

Coordination of Circulation and Gas Exchange

To appreciate how the gas exchange and circulatory systems function together, let's track the variation in partial pressure for O_2 and CO_2 across these systems **(Figure 42.29)**. **1** During inhalation, fresh air mixes with air remaining in the lungs. **2** The resulting mixture formed in the alveoli has a higher P_{O_2} than the blood flowing through the alveolar capillaries. Consequently, there is a net diffusion of O_2 down its partial pressure gradient from the air in the alveoli to the blood. Meanwhile, the presence of a P_{CO_2} in the alveoli that is higher in the capillaries than in the air drives the net diffusion of CO_2 from blood to air. **3** By the time the blood leaves the lungs in the pulmonary veins, its P_{O_2} and P_{CO_2} match the values for the air in alveoli. After returning to the heart, this blood is pumped through the systemic circuit.

Figure 42.29

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Loading and unloading of respiratory gases.
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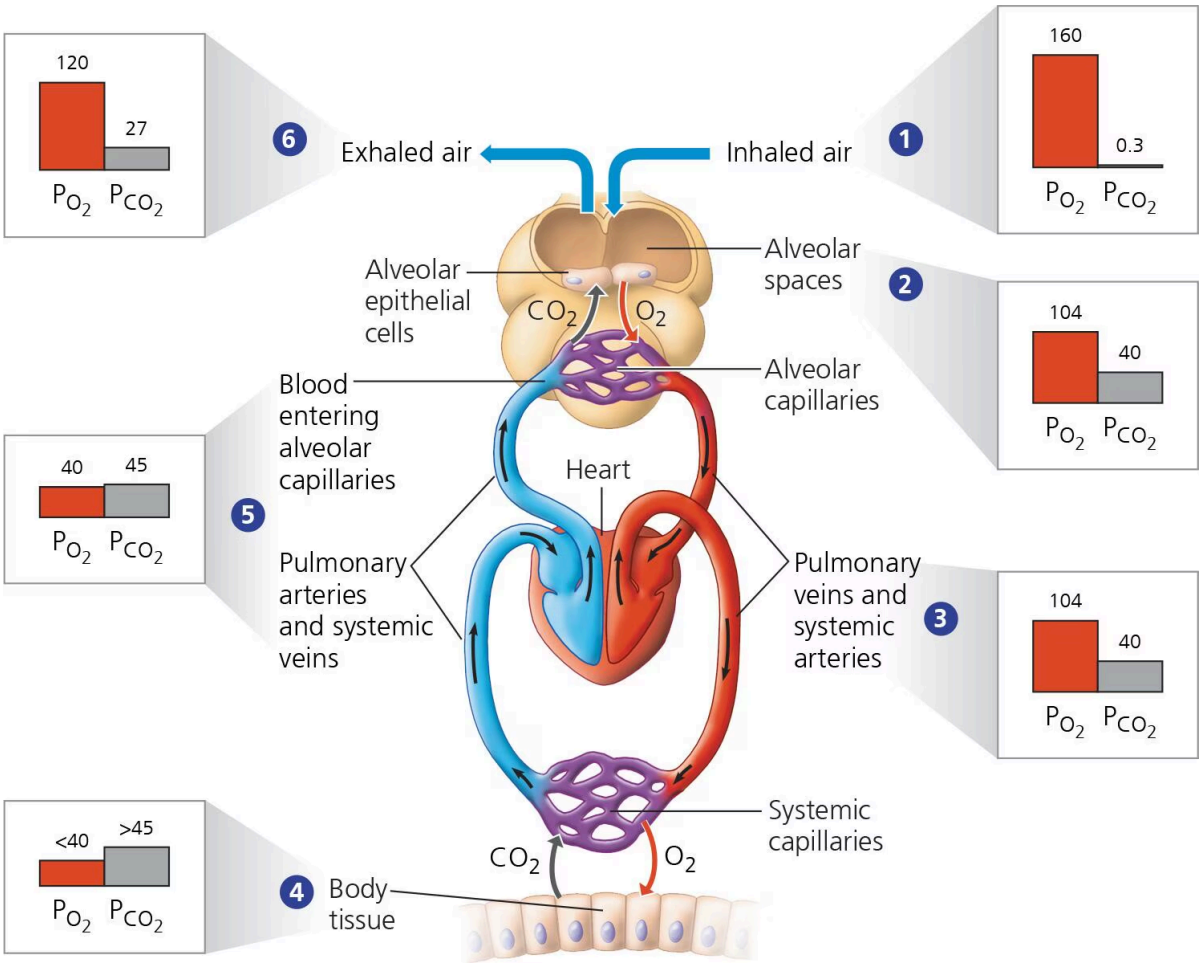


Gas exchange is diagrammed in the respiratory circuit, including partial pressures for O 2 and C O 2. Units for partial pressures= millimeters of mercury. The process is as follows.

1. Air is inhaled into the lungs. Partial pressure of O 2 is 160 and partial pressure of C O 2 is 0.2.
2. In the alveolar spaces, partial pressure of O 2 is 104 and C O 2 is 40. O 2 diffuses into the blood in the alveolar capillaries and C O 2 diffuses out of the capillaries.
3. In the oxygen-rich blood of the pulmonary veins (leading from lungs to left side of heart) and the systemic arteries (leading from left side of heart to systemic circuit) partial pressure of O2 is 104 and partial pressure of C O 2 is 40 (note: these are the same partial pressures as seen for the gasses in the alveoli).
4. As oxygen-rich blood passes from the system arteries over the system capillaries, oxygen diffuses out of the blood and into the body tissue cells, while C O 2 diffuses out of the body tissue cells and into the blood. Partial pressure of O 2 is less than 40 and for C O 2 is greater than 45.

5. Oxygen-poor blood leaves the systemic capillaries and travels towards the heart in the systemic veins. It enters the right atrium of the heart, and then the right ventricle, where it is pumped towards the lungs in the pulmonary arteries, eventually reaching the alveolar capillaries. Partial pressure of oxygen for this stretch of the circuit is 40 and for C O 2 is 45.
6. From the alveoli in the lungs, C O 2 is exhaled. Partial pressure of O 2 is 120 and of C O 2 is 27.

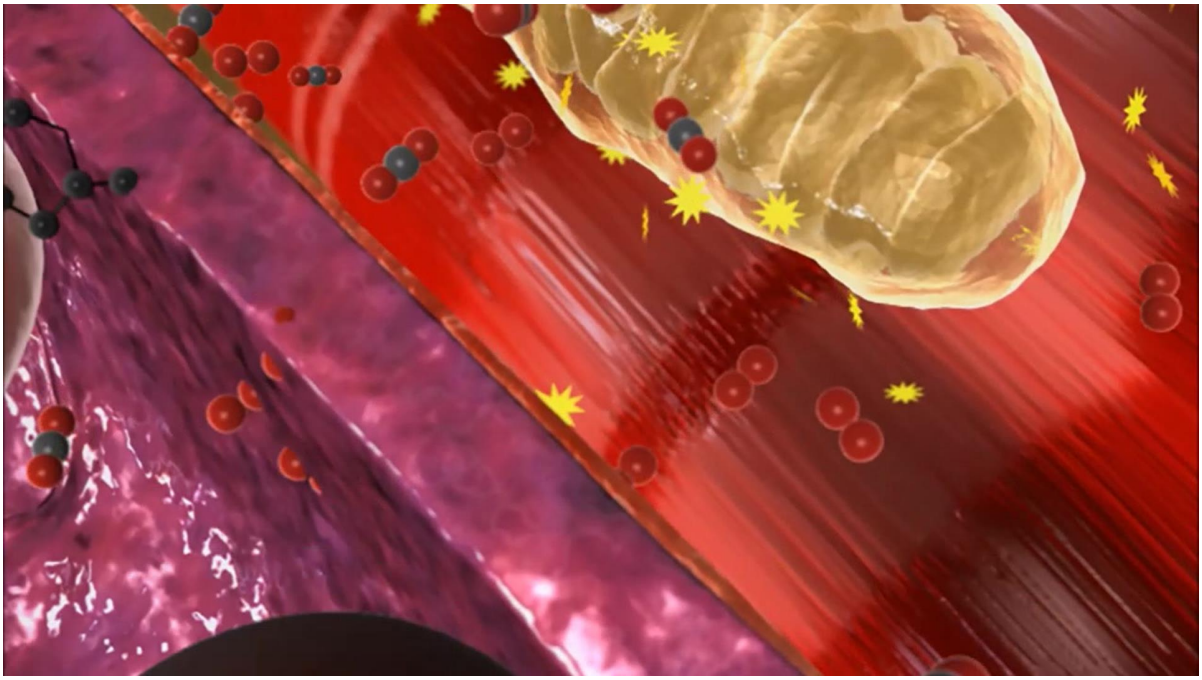
The partial pressures of O_2 and CO_2 are given in mm Hg.

WHAT IF? *If you consciously forced more air out of your lungs each time you exhaled, how would that affect the values shown in the figure?*

*For suggested answer, see*Appendix A*.*

BioFlix® Animation

Gas Exchange in the Human Body



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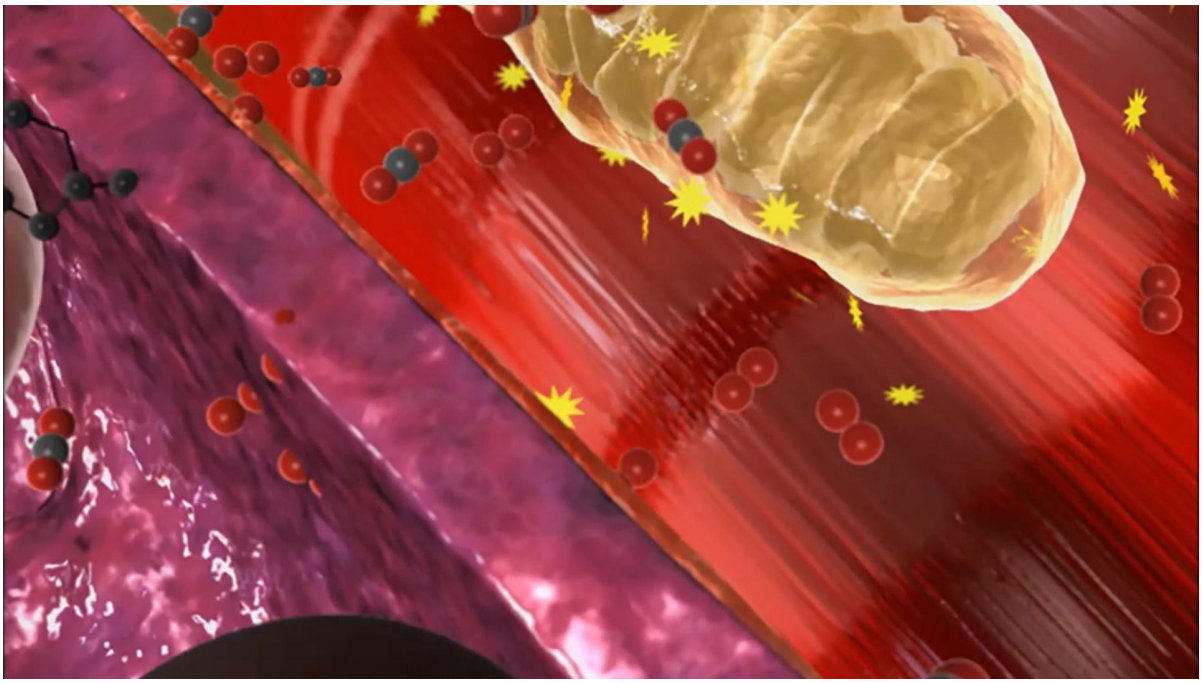

4 In the systemic capillaries, gradients of partial pressure favor the net diffusion of O_2 out of the blood and CO_2 into the blood. These gradients exist because cellular respiration in the mitochondria of cells near each capillary removes O_2 from and adds CO_2 to the surrounding interstitial fluid. **5** Having unloaded O_2 and loaded CO_2, the blood is returned to the heart and pumped to the lungs again. **6** There, exchange occurs across the alveolar capillaries, resulting in exhaled air enriched in CO_2 and partially depleted of O_2.

Respiratory Pigments

The low solubility of O_2 in water (and thus in blood) poses a problem for animals that rely on the circulatory system to deliver O_2. For example, a person requires almost 2 L of O_2 per minute during intense exercise, and all of it must be carried in the blood from the lungs to the active tissues. At normal body temperature and air pressure, however, only 4.5 mL of O_2 can dissolve into a liter of blood in the lungs. Even if 80% of the dissolved O_2 were delivered to the tissues, the heart would still need to pump 555 L of blood per minute!

In fact, animals transport most of their O_2 bound to proteins called **respiratory pigments**. Respiratory pigments circulate with the blood or hemolymph and are often contained within specialized cells. The pigments greatly increase the amount of O_2 that can be carried in the circulatory fluid (from 4.5 to about 200 mL of O_2 per liter in mammalian blood). In our example of an exercising human with an O_2 delivery rate of 80%, the presence of a respiratory pigment reduces the cardiac output necessary for O_2 transport to a manageable 12.5 L of blood per minute.

A variety of respiratory pigments have evolved in animals. With a few exceptions, these molecules have a distinctive color (hence the term *pigment*) and consist of a metal bound to a protein. One example is the blue pigment *hemocyanin*, which has copper as its oxygen-binding component and is found in arthropods and many molluscs.

The respiratory pigment of many invertebrate species and almost all species of vertebrates is hemoglobin (see Concept 42.4). In vertebrates, it is contained in erythrocytes and has four subunits. Each consists of a polypeptide and a heme group, a cofactor that has an iron atom at its center **(Figure 42.30)**. Each iron atom binds one molecule of O2, so a hemoglobin molecule can carry four O2 molecules. Like all respiratory pigments, hemoglobin binds O2 reversibly, loading O2 in the lungs or gills and unloading it elsewhere in the body. This process is enhanced by cooperativity between the hemoglobin subunits (see Concept 8.5). When O2 binds to one subunit, the others change shape slightly, increasing affinity for O2. When four O2 molecules are bound and one subunit unloads its O2, the other three subunits more readily unload O2, as an associated change in shape lowers their affinity for O2.

Figure 42.30

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Ribbon model of hemoglobin.
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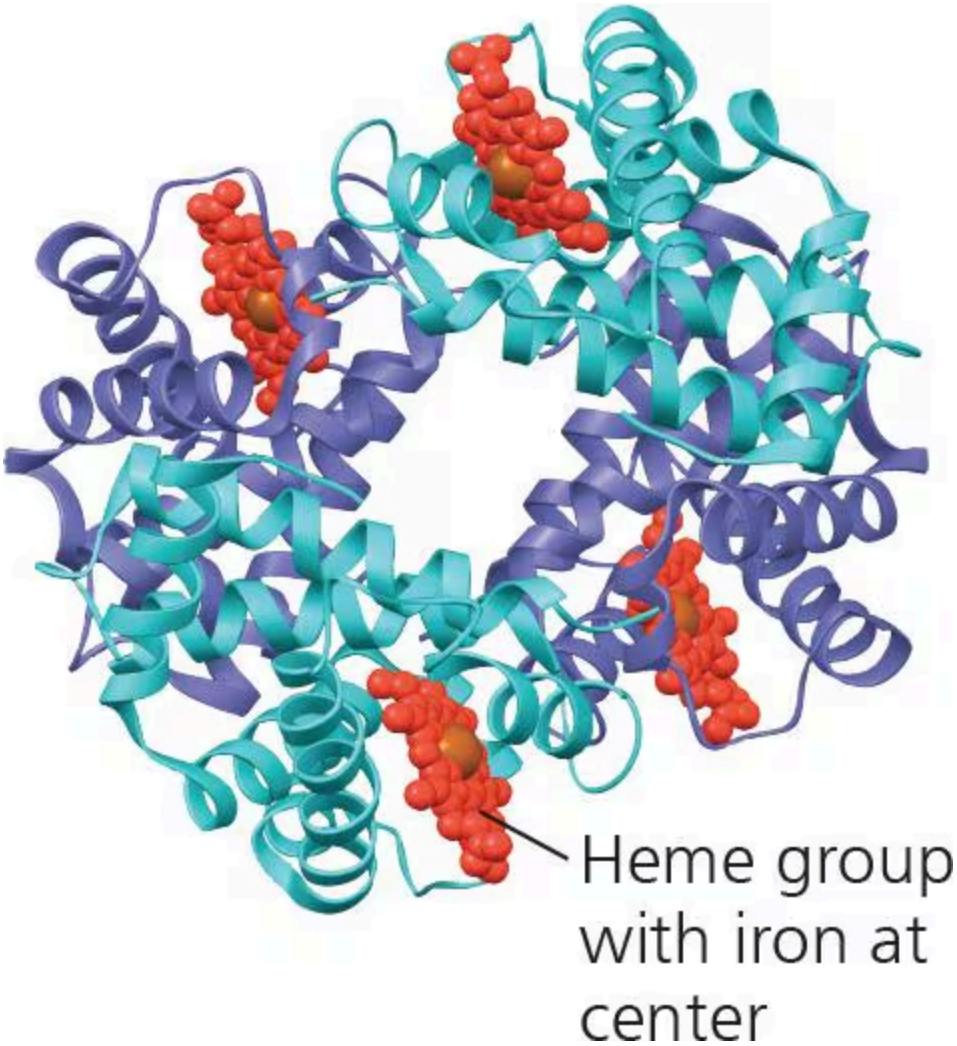
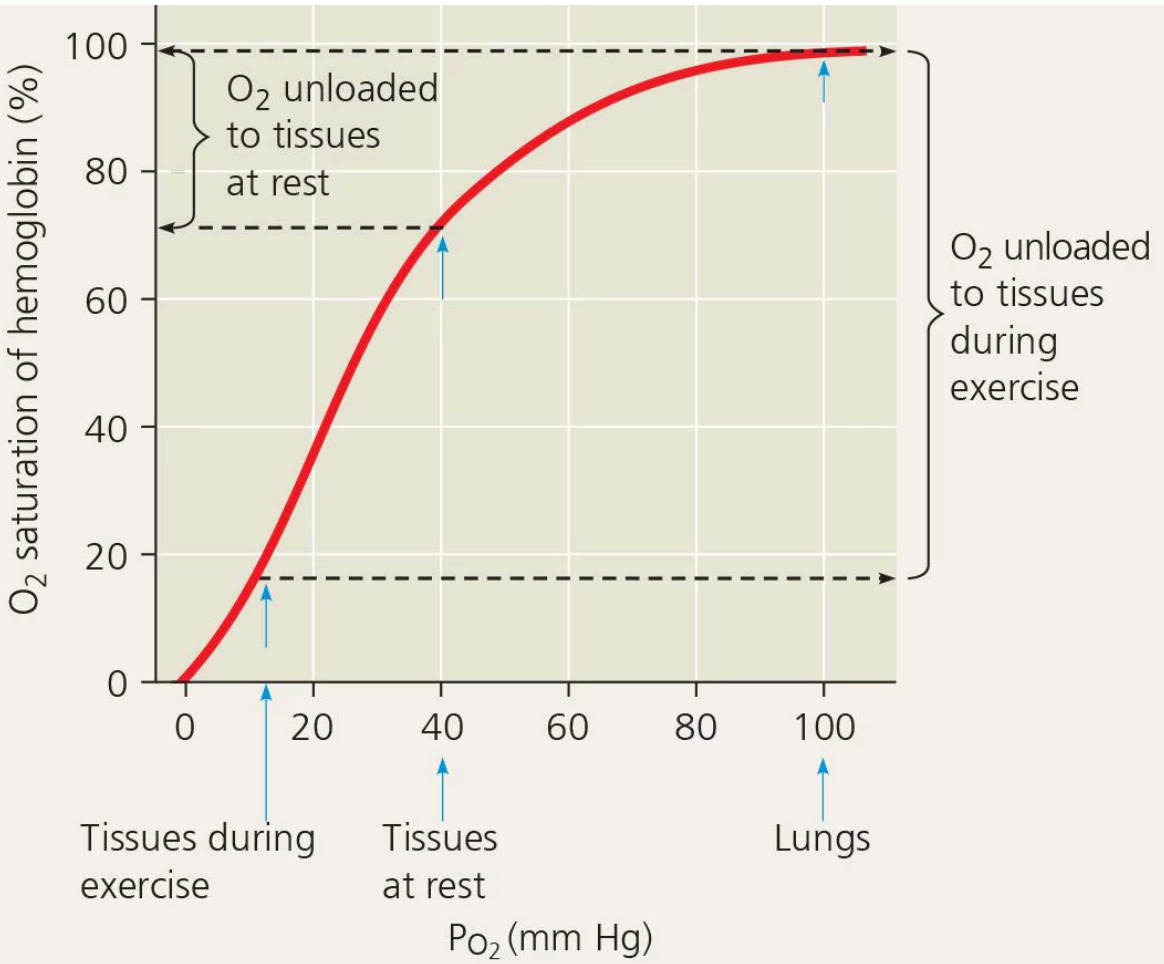


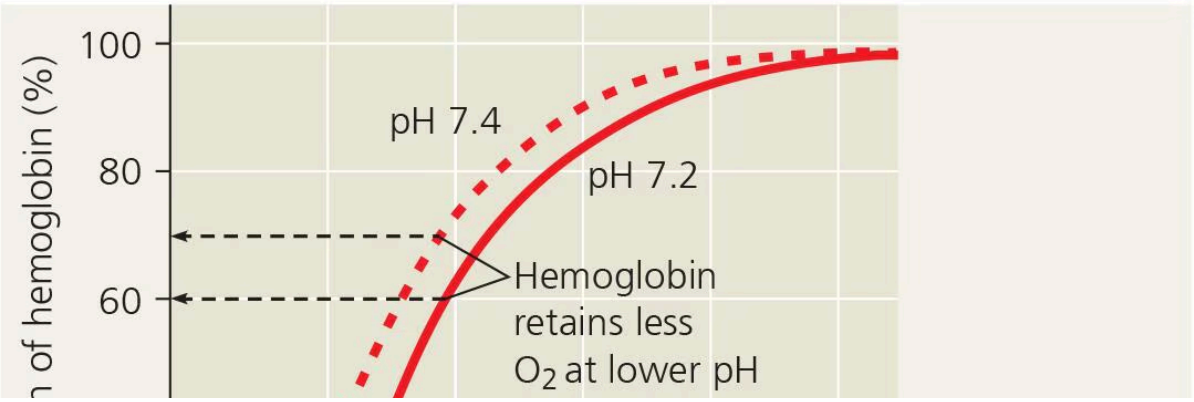
Cooperativity in O_2 binding and release is evident in the dissociation curve for hemoglobin **(Figure 42.31a)**. Over the range of PO_2 where the dissociation curve has a steep slope, even a slight change in PO_2 causes hemoglobin to load or unload a substantial amount of O_2. The steep part of the curve corresponds to the range of PO_2 found in body tissues. When cells in a particular location begin working harder—during exercise, for instance—PO_2 dips in their vicinity as the O_2 is consumed in cellular respiration. Because of subunit cooperativity, a slight drop in PO_2 causes a relatively large increase in the amount of O_2 the blood unloads.

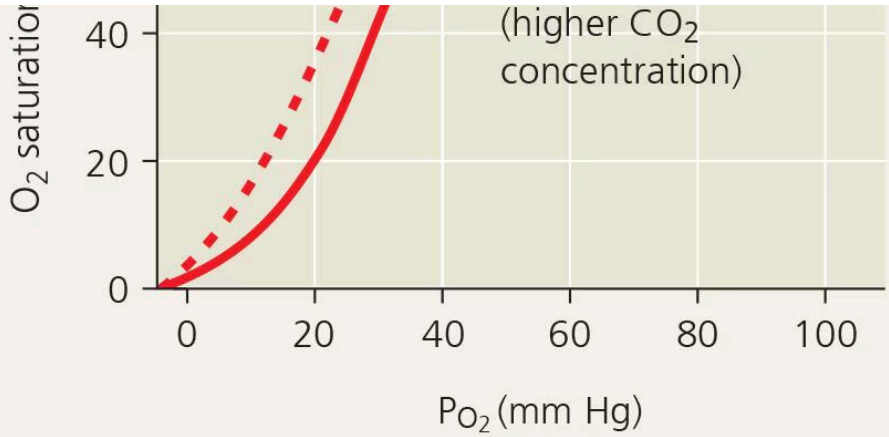
Figure 42.31

Dissociation curves for hemoglobin at 37°C.



(a) P_{O_2} and hemoglobin dissociation at pH 7.4. The curve shows the relative amounts of O_2 bound to hemoglobin exposed to solutions with different P_{O_2}. At a P_{O_2} of 100 mm Hg, typical in the lungs, hemoglobin is about 98% saturated with O_2. At a P_{O_2} of 40 mm Hg, common in resting tissues, hemoglobin is about 70% saturated, having unloaded nearly a third of its O_2. As shown in the above graph, hemoglobin can release much more O_2 to metabolically very active tissues, such as muscle tissue during exercise.





(b) pH and hemoglobin dissociation. In very active tissues, CO_2 from cellular respiration reacts with water to form carbonic acid, decreasing pH. Because hydrogen ions affect hemoglobin shape, a drop in pH shifts the O_2 dissociation curve toward the right (the Bohr shift). For a given P_{O_2}, hemoglobin releases more O_2 at a lower pH, supporting increased cellular respiration.

The graphs are as follows.

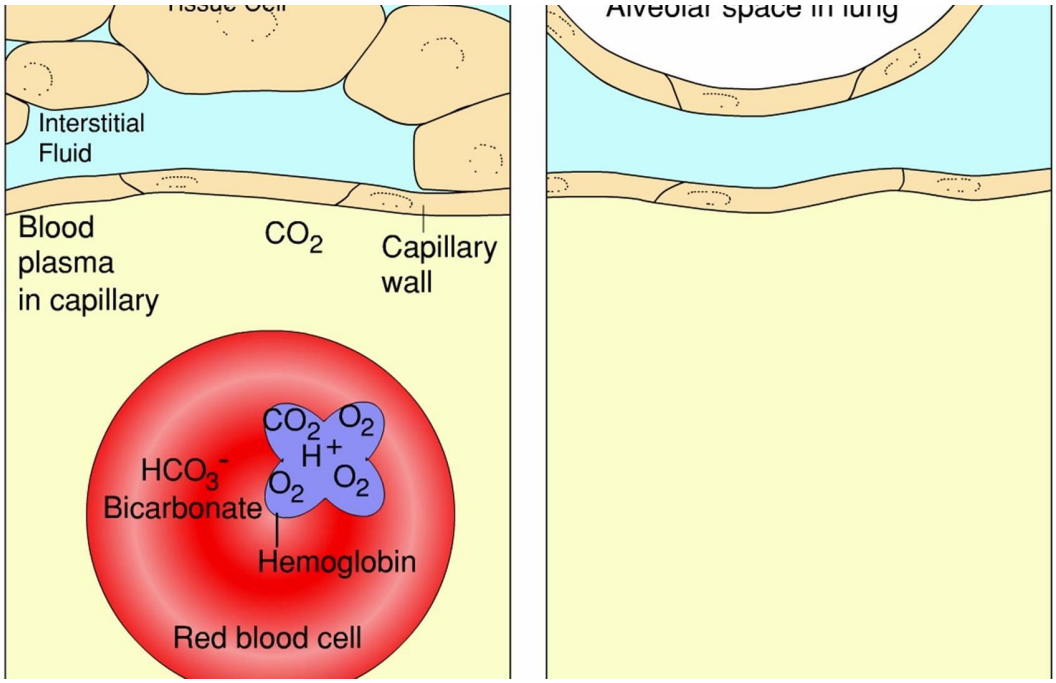
Part a. P O 2 and hemoglobin dissociation at p H 7.4. This dissociation curve shows the relative amounts of O 2 bound to hemoglobin exposed to solutions with different P O 2. At a P O 2 of 100 mm Hg, typical in the lungs, hemoglobin is about 98% saturated with O 2 . At a P O 2 of 40 millimeters Hg, common in resting tissues, hemoglobin is about 70% saturated, nearly a third of its O 2 having been released by dissociation. As shown in the above graph, hemoglobin can release much more O 2 to metabolically very active tissues, such as muscle tissue during exercise. Graph of O 2 saturation of hemoglobin versus partial pressure of oxygen in tissues during exercise, tissues at rest, and in the lungs. The x-axis is partial pressure of oxygen in milligrams of mercury, ranging from 0 to 100 with increments every 20. The y-axis is O 2 saturation of hemoglobin (percent), ranging from 0 to 100 with increments every 20. The line climbs from 0 to 20% saturation by a partial pressure of oxygen of about 17 milligrams of mercury, which represents tissues during exercise. From that point to where the line flattens out (at an O 2 saturation of 100% and a partial pressure for oxygen of about 80 milligrams of mercury) O 2 is unloaded to tissues during exercise. From 70 to 100% O 2 saturation, partial pressure of O 2 is 40 to 80 milligrams of mercury and O 2 is unloaded to tissues at rest. In the lungs, partial pressure of O 2 is 100 milligrams of mercury and O 2 saturation of hemoglobin is 100%.

Part b. p H and hemoglobin dissociation. In very active tissues, C O 2 from cellular respiration reacts with water to form carbonic acid, decreasing pH. Because hydrogen ions affect hemoglobin shape, a drop in p H shifts the O 2 dissociation curve toward the right (the Bohr shift). For a given P O 2, hemoglobin releases more O 2 at a lower pH, supporting increased cellular respiration. Graph of O 2 saturation of hemoglobin versus partial pressure of oxygen at pH 7.4 and pH 7.2 (two separate lines). The x-axis is partial pressure of

oxygen in milligrams of mercury, ranging from 0 to 100 with increments every 20. The y-axis is O 2 saturation of hemoglobin (percent), ranging from 0 to 100 with increments every 20 percent. The line for p H 7.4 is the same as the line in Figure 42.30a. The line for p H 7.2 is roughly parallel to the 7.4 line, but shifted slightly to the right (the Bohr shift). Hemoglobin retains less O 2 at lower p H (higher C O 2 concentration); for example, at p H 7.2 there is 60% O 2 saturation at 40 milligrams of mercury, but at p H 7.4 there is 70% O 2 saturation at 40 milligrams of mercury. Both lines converge at about 70 milligrams of mercury and an O 2 saturation of hemoglobin of about 90%.

Animation

Transport of Respiratory Gases



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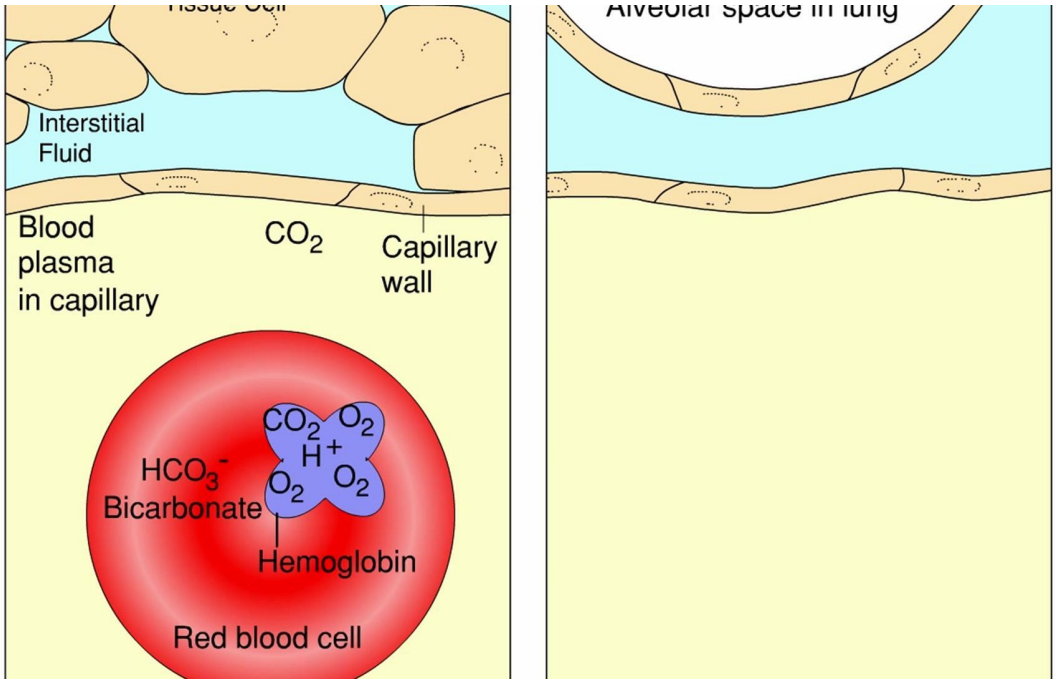

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Hemoglobin is especially efficient at delivering O2 to tissues actively consuming O2. However, this increased efficiency results not from O2 consumption, but rather from CO2 production. As tissues consume O2 in cell respiration, they also produce CO2. As we have seen, CO2 reacts with water, forming carbonic acid, which lowers the pH of its surroundings. Low pH decreases the affinity of hemoglobin for O2, an effect called the **Bohr shift(Figure 42.31b)**. Thus, where CO2 production is greater, hemoglobin releases more O2, which can then be used to support more cellular respiration.

Hemoglobin also assists in buffering the blood—that is, preventing harmful changes in pH. In addition, it has a minor role in CO2 transport, the topic we'll explore next.

Carbon Dioxide Transport

Only about 7% of the CO2 released by respiring cells is transported in solution in blood plasma. The rest diffuses from plasma into erythrocytes and reacts with water (assisted by the enzyme carbonic anhydrase), forming H2CO3. The H2CO3 readily dissociates into H+ and HCO3−. Most H+ binds to hemoglobin and other proteins, minimizing change in blood pH. Most HCO3− diffuses out of the erythrocytes and is transported to the lungs in the plasma. The remaining HCO3−, representing about 5% of the CO2, binds to hemoglobin and is transported in erythrocytes.

When blood flows through the lungs, the relative partial pressures of CO2 favor the net diffusion of CO2 out of the blood. As CO2 diffuses into alveoli, the amount of CO2 in the blood decreases. This decrease shifts the chemical equilibrium in favor of the conversion of HCO3− to CO2, enabling further net diffusion of CO2 into

alveoli. Overall, the PCO_2 gradient is sufficient to drive a 10–15% reduction in PCO_2 during passage of blood through the lungs (see Figure 42.29).

Respiratory Adaptations of Diving Mammals

EVOLUTION Animals vary greatly in their ability to spend time in environments in which there is no access to their normal respiratory medium—for example, when an air-breathing mammal swims underwater. Whereas most humans cannot swim deeper than 20m or hold their breath longer than 2–3 minutes, the Weddell seal of Antarctica routinely plunges to 200–500 m and remains there for a period ranging from 20 minutes to more than an hour **(Figure 42.32)**. Another diving mammal, the Cuvier's beaked whale, can reach depths of 2,900 m—nearly 2 miles—and stay submerged for more than 2 hours! What enables these amazing feats?

Figure 42.32

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A Weddell seal, adapted to deep, long dives.
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One evolutionary adaptation of diving mammals to prolonged stays underwater is a capacity to store large amounts of O_2 in their bodies. The volume of blood per kilogram of body mass in a Weddell seal is about twice that in a human. Furthermore, the muscles of seals and other diving mammals contain a high concentration of an oxygen-storing protein called **myoglobin** in their muscles. As a result, the Weddell seal can store about twice as much O_2 per kilogram of body mass as can a human.

Diving mammals not only have a relatively large O_2 stockpile but also have adaptations that conserve O_2. They swim with little muscular effort and glide passively for prolonged periods. During a dive, their heart rate and O_2 consumption rate decrease, and most blood is routed to vital tissues: the brain, spinal cord, eyes, adrenal glands, and, in pregnant seals, the placenta. Blood supply to the muscles is restricted or, during extended dives, shut off altogether. During these dives, a Weddell seal's muscles deplete the O_2 stored in myoglobin and derive most of their ATP from anaerobic respiration (see Concept 9.5).

How might such adaptations have arisen over the course of evolution? All mammals, including humans, have a diving reflex triggered by a plunge or fall into water: When the face contacts cold water, the heart rate immediately decreases and blood flow to body extremities is reduced. Genetic changes that strengthened this reflex would have provided a selective advantage to seal ancestors foraging underwater. Also, genetic variations that increased traits such as blood volume or myoglobin concentration would have improved diving ability and therefore been favored during selection over many generations.